

Higher Math Elective HW1

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October 3, 2026

1. Using the truth tables we've discussed in class, evaluate the following expressions in classical Boolean logic with two truth values $B = \{T, F\}$:

(a) $(T \vee F) \Rightarrow F = F$;

(b) $(T \vee F) \Rightarrow T = T$;

(c) $(T \wedge F) \Rightarrow F = T$;

(d) $(T \wedge F) \Rightarrow F = T$;

(e) $T \vee (F \Rightarrow F) = T \vee T = T$;

(f) $T \vee (F \Rightarrow T) = T \vee T = T$;

(g) $T \wedge (F \Rightarrow F) = T \wedge T = T$;

(h) $T \wedge (F \Rightarrow F) = T \wedge T = T$;

(i) $(F \vee ((F \Rightarrow T) \Rightarrow T)) \Leftrightarrow (((T \Rightarrow F) \Rightarrow T) \wedge T)$.

$$F \vee T \Leftrightarrow T \wedge T$$

$$T \Leftrightarrow T = T$$

2. Construct the truth tables for the following predicates:

(a) $P(x) = x \vee \neg x$

x	$x \vee \neg x$
T	T
F	T

(b) $P(x) = x \Rightarrow \neg x$

x	$x \Rightarrow \neg x$
T	F
F	T

(c) $P(x, y) = x \Rightarrow \neg y$

x	y	$x \Rightarrow \neg y$
T	T	F
T	F	T
F	T	T
F	F	T

(d) $P(x, y) = y \Rightarrow \neg x$

x	y	$y \Rightarrow \neg x$
T	T	F
T	F	T
F	T	T
F	F	T

(e) $P(x, y, z) = (x \iff y) \wedge (y \iff z)$

x	y	z	$P(x, y, z)$
T	F	F	F
T	F	T	F
T	T	F	F
T	T	T	T
F	F	F	T
F	F	T	F
F	T	F	F
F	T	T	F

(f) $P(x, y, z) = (x \iff y) \wedge (y \iff z)$

x	y	z	$P(x, y, z)$
T	F	F	F
T	F	T	F
T	T	F	F
T	T	T	T
F	F	F	T
F	F	T	F
F	T	F	F
F	T	T	F

(g) $P(x, y, z) = (x \iff y) \wedge (y \iff z) \wedge (y \neq z)$

x	y	z	$P(x, y, z)$
T	F	F	F
T	F	T	F
T	T	F	F
T	T	T	F
F	F	F	F
F	F	T	F
F	T	F	F
F	T	T	F

(h) $P(x, y, z) = (x \neq y) \vee (y \neq z)$

x	y	z	$P(x, y, z)$
T	F	F	T
T	F	T	T
T	T	F	T
T	T	T	F
F	F	F	F
F	F	T	T
F	T	F	T
F	T	T	T

3. **Theorem 1** (De Morgan's "laws"). *Given two statements P and Q , the following equalities hold:*

$$\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q),$$

$$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q).$$

Proof. It's your task to prove this.

Will use truth table for all possible values of P,T to prove the statements

P	Q	$\neg(P \vee Q)$	$(\neg P) \wedge (\neg Q)$	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$
T	T	F	F	F	F
T	F	F	F	T	T
F	T	F	F	T	T
F	F	T	T	T	T

Thus, for $P = \{T, F\}$ and $Q = \{T, F\}$, both statement hold

4. **Theorem 2** (De Morgan's "laws" for sets, universal form). *Let U be a universe. For any sets $A, B \subseteq U$ (where c denotes complement relative to U),*

$$(A \cup B)^c = A^c \cap B^c, \quad (1)$$

$$(A \cap B)^c = A^c \cup B^c. \quad (2)$$

Proof. Using Theorem 1.

LHS1. Suppose $x \in (A \cup B)^c$. Then need to prove that $(A \cup B)^c \subseteq A^c \cap B^c$.

Proof:

$$x \in (A \cup B)^c \Rightarrow \neg(x \in A \cup B) \Rightarrow \neg(x \in A \vee x \in B) \xrightarrow{\text{Theorem 1}} \neg(x \in A) \wedge \neg(x \in B) \Rightarrow x \in A^c \cap B^c.$$

Proven.

RHS1. Suppose $x \in A^c \cap B^c$. Then need to prove that $A^c \cap B^c \subseteq (A \cup B)^c$.

Proof:

$$x \in A^c \cap B^c \Rightarrow \neg(x \in A) \wedge \neg(x \in B) \xrightarrow{\text{Theorem 1}} \neg(x \in A \vee x \in B) \Rightarrow \neg(x \in A \cup B) \Rightarrow x \in (A \cup B)^c.$$

Proven.

Then statement $(A \cup B)^c = A^c \cap B^c$ holds.

LHS2. Suppose $x \in (A \cap B)^c$. Then need to prove that $(A \cap B)^c \subseteq A^c \cup B^c$.

Proof:

$$x \in (A \cap B)^c \Rightarrow \neg(x \in A \cap B) \Rightarrow \neg(x \in A \wedge x \in B) \xrightarrow{\text{Theorem 1}} \neg(x \in A) \vee \neg(x \in B) \Rightarrow x \in A^c \cup B^c.$$

Proven.

RHS2. Suppose $x \in A^c \cup B^c$. Then need to prove that $A^c \cup B^c \subseteq (A \cap B)^c$.

Proof:

$$x \in A^c \cup B^c \Rightarrow \neg(x \in A) \vee \neg(x \in B) \xrightarrow{\text{Theorem 1}} \neg(x \in A \wedge x \in B) \Rightarrow \neg(x \in A \cap B) \Rightarrow x \in (A \cap B)^c.$$

Proven.

Then statement $(A \cap B)^c = A^c \cup B^c$ also holds.

5. **Theorem 3** (De Morgan's "laws" for sets, general form). *For any sets A, B, C ,*

$$\begin{aligned} A \setminus (B \cup C) &= (A \setminus B) \cap (A \setminus C), \\ A \setminus (B \cap C) &= (A \setminus B) \cup (A \setminus C). \end{aligned}$$

Proof. Prove using Theorem 1.

LHS1. Let $\Phi x \in A \setminus (B \cup C)$. Then $x \in A \wedge x \notin (B \cup C) \Rightarrow x \in A \wedge x \notin B \wedge x \notin C \Rightarrow x \in (A \setminus B) \cap (A \setminus C)$. Proven

RHS1. Let $\Phi x \in (A \setminus B) \cap (A \setminus C)$. Then $(x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C) \Rightarrow x \in A \wedge x \notin B \wedge x \notin C \Rightarrow x \in A \wedge x \notin (B \cup C) \Rightarrow x \in (A \setminus (B \cup C))$. Proven

Then statement $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ holds.

LHS2. Let $\Phi x \in A \setminus (B \cap C)$. Then $x \in A \wedge x \notin (B \cap C) \Rightarrow x \in A \wedge (x \notin B \vee x \notin C) \Rightarrow (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \Rightarrow x \in ((A \setminus B) \cup (A \setminus C))$. Proven

RHS2.

Let $\Phi x \in ((A \setminus B) \cup (A \setminus C))$. Then $(x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \Rightarrow x \in A \wedge (x \notin B \vee x \notin C) \Rightarrow x \in A \wedge x \notin (B \cap C) \Rightarrow x \in (A \setminus (B \cap C))$. Proven

Then statement $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ also holds.

6. **Definition 4.** *Symmetrical difference* of two sets A and B is defined as a set of all elements that belong exactly to one of these two sets and is denoted as $A \Delta B$.

Your task: Express the symmetrical difference in at least two different ways using all previously-defined operations ($\cup, \cap, \setminus$). Also define it using abstraction method (it will look like $A \Delta B = \{x \mid \dots\}$).

$$A \Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B) = \{x \mid (x \in A \wedge x \notin B) \vee (x \notin A \wedge x \in B)\}$$